Choosing Training Data Edits for Safer Language Models

Research proposal for discussion | September 2026

Question. Given a measurable safety failure and a fixed menu of post-training data removals or replacements, can we identify the edit that most improves safety on held-out examples, subject to limits on utility loss and over-refusal, without fully retraining every candidate?

Motivation and prior work. SOURCE predicts data-removal effects through approximate unrolled differentiation [1]; IF-Guide already uses influence-based interventions to reduce toxicity [2]; distributional attribution studies effects across stochastic training runs [3]. I want to investigate when these tools support a reliable intervention decision. Finite group edits can invalidate local approximations, and uncertainty across training seeds does not capture systematic prediction error. I propose testing whether attribution-specific error signals improve how validation compute is allocated. A useful result would require better edit choices at matched total compute; novelty remains to be established.

Working hypothesis and approach. Disagreement between influence-based predictions and short training continuations may identify edits whose apparent safety benefit or utility cost is unreliable. I propose using these signals to allocate a limited number of complete post-training runs, correcting candidate estimates as evidence arrives. The selector would distinguish training variability from surrogate error, retain the unchanged-data option, and report when its budget cannot resolve the decision. The decisive comparison is the same selector without influence scores, given the same calibration edits and total compute budget. I would also measure how often both predictors agree yet misrank unseen edits.

First experiment. I would start with one small open-weight model, a fixed supervised fine-tuning recipe, and one prespecified failure mode, such as toxic responses. I would freeze the candidate menu, utility and over-refusal limits before evaluation. Edits would span group removals and replacements at different sizes. Runs would restart from the same pre-SFT checkpoint, with a fixed schedule and treatment of removed tokens. Randomly sampled calibration edits and evaluation edits would use separate underlying data groups. Selectors could access validation prompts; final comparisons would use untouched audit prompts and fresh training seeds after each selector commits to an edit. Blinded checks of sampled outputs would audit the safety scorer.

Comparisons and success criteria. Selection baselines would include random and toxicity-based rankings, influence/SOURCE rankings, uniform or adaptive retraining, and generic multi-fidelity selection using short and full runs. Selectors would share a candidate menu and compute budget; IF-Guide would provide a separate end-to-end toxicity baseline. An exhaustive, repeated-retraining study of the small menu would provide an uncertain reference for the best feasible edit. I would measure the safety gap to that reference, utility and over-refusal constraint violations, uncertainty calibration and total compute. Scoring, calibration, short runs and final validation would all count; shared reference-study costs would be reported separately and its results withheld from selectors. Success requires a better decision-quality-versus-compute trade-off than the strongest baseline. If selective validation adds no benefit after overhead, the proposed method is unsupported.

Preparation and guidance sought. I have completed a separate toy pilot on occurrence-level attribution and variance reduction; it did not establish data-edit selection or practical savings. I am now scoping this study. I would value your view on whether the finite-edit error problem offers a substantive gap beyond existing attribution and adaptive evaluation methods, and which smallest experiment would most convincingly test it.

Selected references

[1] Bae et al. (2024). Training Data Attribution via Approximate Unrolled Differentiation.

[2] Coalson et al. (2025). IF-Guide: Influence Function-Guided Detoxification of LLMs.

[3] Mlodozeniec et al. (2025). Distributional Training Data Attribution: What do Influence Functions Sample?